5TH GRADE MATH: EQUIVALENT FRACTIONS

WHAT ARE EQUIVALENT FRACTIONS?

Definition: Equivalent fractions are fractions that name the exact same amount of a whole, even though they use different numerators and denominators.

1. The Pizza Model (Visual Proof)

Imagine you and a friend order two identical pizzas:

Pizza A is cut into 2 equal slices. You eat 1 slice (1/2).
Pizza B is cut into 4 equal slices. Your friend eats 2 slices (2/4).



Both of you ate the **exact same** amount of pizza! Therefore, 1/2=2/4.

2. The Calculator Check (Decimal Verification)

Fraction	Calculator Division	Decimal Value	Are they equivalent?
1/2	1 ÷ 2	0.5	YES
2/4	2 ÷ 4	0.5	YES
3/6	3 ÷ 6	0.5	YES

Rule: Because every division result equals 0.5, all three fractions are 100% equivalent!

3. How to Create and Test Equivalent Fractions

MATH RULES FOR EQUIVALENT FRACTIONS

Multiplying / Dividing Rule: Multiply or divide both numerator and denominator by the same non-zero number:

$$\frac{1}{2} = \frac{1 \times 2}{2 \times 2} = \frac{2}{4}$$

Student Activity: Equivalent Fraction Sorting Challenge

Instructions: 1. Simplify each fraction in the Fraction Bank or use a calculator to find its decimal value.
2. Find the equivalent fractions for each target group.
3. Write each fraction into its designated **Set Box** below!

FRACTION BANK (Choose from these 6 fractions): 1/3 2/5 2/6 4/10 3/9 6/15

SET 1: Fractions Equivalent to 1/3
(Decimal Value: 0.333...)

1. _____
2. _____
3. _____

SET 2: Fractions Equivalent to 2/5
(Decimal Value: 0.4)

1. _____
2. _____
3. _____